

M-ENDPOINT evaluation with a convergence certificate and single-J0 conservatism

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-07 · this version issued 2026-06-07 · v1.1

1. Purpose and scope

SC-SCOPE's all-orders lift is gated by four registered inputs (M-ENDPOINT, GHAT3-Q0 optional, GHAT4-PERTRANSFER, R-U6-1). This note resolves the first: M-ENDPOINT = $M(\hat{r}(2 \times 10^{-3})) = M(0.33675)$, the convexity-honest endpoint dressing variance. The earlier sunset-endpoint estimate used a linear-response slope $M'(\hat{r}) = -J(0)$ (the correct relation is $-J(0)/2$), invalidating the $\times 1.34$ figure; here the constant is evaluated DIRECTLY by quadrature, so the linearisation does not enter. **Scope: the SUNSET axis only.** This note does not close SC-SCOPE.

2. The M-ENDPOINT constant with a convergence certificate

With $M(m) = \frac{1}{2\pi^2} \int_0^\infty k^2 / [m + C(k^2 - q_0^2)^2] dk$,

$$\boxed{M\text{-ENDPOINT} = M(0.33675) = 0.104953} \quad (M_R = 0.109414).$$

Provenance / certificate. The production value (M_fast) carries an analytic tail model. An independent radial trapezoid (cutoff $k_{\max} = 60$, $n = 10^6$ points) gives 0.105589, agreeing to 0.61%; the residual is resonance-peak ($k \sim q_0$) quadrature resolution plus tail treatment, not a value error. The analytic truncation beyond $k_{\max}$ is bounded by $\int_{k_{\max}}^\infty \frac{dk}{2\pi^2 C k^2} = \frac{1}{2\pi^2 C k_{\max}} = 8.4 \times 10^{-4}$. M-ENDPOINT is therefore an EXECUTED constant cross-checked at the $\sim 1\%$ level (raw multi-resolution envelope 8.5×10^{-4}); it is NOT claimed as a 0.1% interval constant. The downstream verdict is certified robust to this envelope (§3). $M(0.33675) < M_R$ as monotonicity requires, and lies in the corrected bracket $[0.10474, 0.10941]$ (the old $[0.1001, 0.1094]$ used the factor-2 slope). $M(0.32064) = 0.107098$ is the $I = 10^{-3}$ companion.

3. The sunset-axis endpoint sign

$u_{\text{eff}}(M) = u + 10vM$; the matched single- J_0 sunset bound and composed AddE margin are $b(I) = \frac{1}{2}u_{\text{eff}}(M)^2(2I)6J_0M$ and $\text{composed}(I) = \text{MARGIN}(1 - 1/\rho(I))$, with $J_0 = 0.2896$, $\text{MARGIN} = 0.004320$ (DERIVED, §0 of the script), $\rho(2 \times 10^{-3}) = 2.6$.

Single- J_0 conservatism. J decreases with dressing, so for the endpoint intensities $J(\hat{r}(I)) \leq J_0$:

$$J(\hat{r}(10^{-3})) = 0.2764, \quad J(\hat{r}(2 \times 10^{-3})) = 0.2565 \leq J_0 = 0.2896.$$

Using the single anchor J_0 therefore OVER-estimates the bound, i.e. the quoted ratios are conservative; the per-dressing- J ratios are larger.

$$\frac{\text{composed}}{b} \Big|_{2 \times 10^{-3}} : \underbrace{\times 0.970}_{\text{frozen } M_R (=U4)} \rightarrow \boxed{\underbrace{\times 1.13}_{\text{dressed, } J_0}} \rightarrow \underbrace{\times 1.28}_{\text{dressed, per-}J} ,$$

and $\times 3.02$ at $I = 10^{-3}$. **Verdict robustness.** Even at M -ENDPOINT plus the full raw envelope (larger M is the worse direction) the ratio is $\times 1.096 > 1$. So at sign level the SUNSET-AXIS endpoint no longer individually fails: the U4 SUNSET-AXIS marginal failure was a frozen-coupling artefact. The $\times 1.13$ is a SIGN result, NOT a comfortable margin – in the joint inequality the $\sim 13\%$ can be consumed by the other channels.

4. What this resolves and what remains

RESOLVED: M -ENDPOINT (value + certificate) and the sunset axis of the third-cumulant endpoint (positive at sign level, robust to the envelope). OPEN: the quartic-difference channel (GHAT4-PERTRANSFER; sup-kernel grade $\times 1.0$, per-transfer form factor load-bearing) and the tadpole alignment (R-U6-1). The all-orders lift requires these re-assembled as a joint second+third-order inequality. SC-SCOPE remains OPEN; the U4 third-cumulant endpoint is NOT yet shown to be an artefact (only its sunset axis is).

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, 6 code-discipline)

- (α) “*You resolved one channel, not SC-SCOPE.*” CONCEDED and stated: the ledger is M -ENDPOINT RESOLVED, sunset axis POSITIVE, SC-SCOPE OPEN on GHAT4-PERTRANSFER + R-U6-1.
- (β) “ $\times 1.13$ is thin.” VALID: claimed as a SIGN only; §3 shows the sign survives the full quadrature envelope ($\times 1.096$) and the per- J ratio is $\times 1.28$; the joint-inequality consumption is flagged.
- (γ) “ $M(0.33675)$ is an executed value, not interval- certified.” VALID and now explicit: cross-checked at $\sim 1\%$ (0.61% method agreement; 8.4×10^{-4} tail bound); the VERDICT, not a 0.1% constant, is what is certified.
- (δ) “ U_4 reproduction proves only consistency with the old estimator.” CONCEDED: the frozen $\times 0.970$ is a REGRESSION sanity check; the load-bearing content is the direct M -ENDPOINT + dressed formula + the single- J_0 conservatism justification.

Quantitative sanity check (mandatory): magnitude – two quadratures agree to 0.61%; limit-case – frozen- M_R reproduces U4 $\times 0.97$ to $< 1\%$; sign-direction – $M(\hat{r}) < M_R$ and $J(\hat{r}) < J_0$, both lowering the sunset bound (selection-protecting); convergence – analytic tail bound 8.4×10^{-4} and the verdict robust to the full envelope.

6. Result footer

Result ID: B5-BEYOND-LAYER-BOUND / SC-SCOPE sunset axis (M -ENDPOINT)
Precise statement: M -ENDPOINT = $M(0.33675) = 0.104953$ (executed; two

quadratures agree 0.61%; analytic tail bound $8.4e-4$). Single-J0 is conservative ($J(r_{\text{hat}}(I)) \leq J_0$). The sunset-axis endpoint ratio is $x1.13 (J_0) / x1.28$ ($\text{per-J} > 1$, robust to the envelope ($x1.10$ worst)). The SUNSET axis of the third-cumulant endpoint is POSITIVE at sign level.

Scope: Sunset axis only; matched single-J0 convention; second + third cumulant sunset channel at the certified intensities. Does NOT close SC-SCOPE.

Dependencies: sectorb_common (margin_of, J_of_t); production gap_solve / M_fast; AddE rho ratios (cited).

Evidence grade: EXECUTED (scscope_mendpoint_eval.py v1.1, 12/12) + independent-quadrature cross-check + verdict-robustness.

Reproduction command: python codes/vacuum/scscope_mendpoint_eval.py

Expected output: 12/12 PASS; M-ENDPOINT=0.10495; dressed endpoint $x1.13$; verdict robust $x1.10$.

Falsification gate: M-ENDPOINT outside $[0.1047, 0.1094]$; the dressed endpoint ratio ≤ 1 over the envelope.

Tier before / after: No tier/gate flip. M-ENDPOINT gate: value RESOLVED (recommend RESOLVED pending operator sign-off). SC-SCOPE stays OPEN (GHAT4-PERTRANSFER + R-U6-1).

No-overclaim statement: Resolves ONE of three SC-SCOPE channels. Sunset axis only; the U4 THIRD-CUMULANT endpoint is NOT shown to be an artefact. $x1.13$ is a SIGN, not a margin. M-ENDPOINT is an executed value cross-checked at $\sim 1\%$, not a 0.1% interval constant.

Next required action: GHAT4-PERTRANSFER (per-transfer quartic form factor); R-U6-1 (tadpole alignment); then the joint second+third-order endpoint inequality.